

AN ASYMPTOTIC REPRESENTATION OF TRIMMED MEANS WITH APPLICATIONS

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SUMMARY

It is shown that the trimmed mean can be represented asymptotically as an average of i.i.d. random variables. Some applications of this result to sequential procedures are indicated.

Introduction

Let $X_1, X_2, \dots, X_n$ be a sample of i.i.d. random variables with common distribution function (df) F and denote by $X_{1n} \leq X_{2n} \leq \dots \leq X_{nn}$ the corresponding order statistics. Let α and β be positive real numbers with $\alpha + \beta < 1$ and put $r = [\alpha n] + 1$, $s = [n(1 - \beta)]$ and $\gamma = 1 - \alpha - \beta$. Let g_1 and g_2 satisfy $F(g_1) = \alpha$ and $F(g_2) = 1 - \beta$. We will assume that F has at least two derivatives in neighbourhoods of g_1 and g_2 with F'' bounded in these neighbourhoods while $F'(g_1)$ and $F'(g_2)$ are positive.

We call

$$(1) \quad M_n = n^{-1} \gamma^{-1} \sum_{i=r}^s X_{in}$$

the α, β -trimmed mean of the sample. (See e.g. Stigler (1973)). It is our purpose to show that M_n can be represented asymptotically as an average of suitably defined i.i.d. random variables. More specifically, define the function

$$(2) \quad \psi(x) = \begin{cases} g_1/\gamma & \text{for } x < g_1 \\ x/\gamma & \text{for } g_1 \leq x \leq g_2 \\ g_2/\gamma & \text{for } x > g_2 \end{cases}$$

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put

$$(3) \quad Y_i = \psi(X_i) - E\psi(X_i)$$

and let

$$(4) \quad \theta = \gamma^{-1} \int_{g_1}^{g_2} x dF(x).$$

Then we prove the following

Theorem 1

$$(5) \quad M_n - \theta = n^{-1} \sum_{i=1}^n Y_i + O(n^{-1} \log \log n) \text{ a.s.}$$

This result is a strong form of the representation contained in the statements at the bottom of page 1051 and the top of page 1052 of Huber (1972) specialised to the trimmed means. From this it follows immediately that M_n is a strongly consistent estimator of θ ; moreover an iterated logarithm law holds for M_n and it follows that $n^{1/2}(M_n - \theta)$ is asymptotically $N(0, \sigma^2)$ distributed with

$$(6) \quad \sigma^2 = \text{Var } Y_i = \gamma^{-2} \left[\int_{g_1}^{g_2} (x - \theta)^2 dF(x) + \alpha a^2 + \beta b^2 - (\beta b - \alpha a)^2 \right]$$

where

$$(7) \quad a = \theta - g_1$$

$$b = g_2 - \theta.$$

The last result has been proved in various other ways in the literature (See e.g. Stigler (1973) and Bickel (1965)).

If F is symmetric and $\alpha = \beta$ then θ equals the point of symmetry of F and $a = b$ and the formulas above simplify somewhat. We will refer to this as the symmetric case.

In this case it has been proposed to estimate σ^2 by

$$(8) \quad S_n^2 = \gamma^{-2} \left[n^{-1} \sum_{i=r}^s (X_{in} - M_n)^2 + \alpha (X_{rn} - M_n)^2 + \alpha (X_{sn} - M_n)^2 \right]$$

(See e.g. Jaeckel (1971)). We will show that S_n^2 can also be represented asymptotically as an average of i.i.d. random variables namely, if

$$(9) \quad \psi^*(x) = \begin{cases} \gamma^{-2}(g_1^2 - 2\alpha g_1/F'(g_1)) & \text{for } x < g_1 \\ \gamma^{-2}x^2 & \text{for } g_1 \leq x \leq g_2 \\ \gamma^{-2}(g_2^2 + 2\alpha g_2/F'(g_2)) & \text{for } x > g_2 \end{cases}$$

and

$$(10) \quad Y_i^* = \psi^*(X_i) - E\psi^*(X_i),$$

then we prove

Theorem 2

$$(11) \quad S_n^2 - \sigma^2 = n^{-1} \sum_{i=1}^n Y_i^* + o(n^{-3/4} \log n) \text{ a.s.}$$

It follows from this that S_n^2 is strongly consistent for σ^2 and that $n^{1/2}(S_n^2 - \sigma^2)$ is asymptotically $N(0, \tau^2)$ distributed, with

$$(12) \quad \tau^2 = \text{Var } Y_i^* = 2\alpha\gamma^{-4}(a^2 + 2\alpha a/F'(g_1))^2 + \gamma^{-4} \int_{g_1}^{g_2} (x-\theta)^4 dF(x) \\ - [2\alpha\gamma^{-2}(a^2 + 2\alpha a/F'(g_1)) + \gamma^{-2} \int_{g_1}^{g_2} (x-\theta)^2 dF(x)]^2.$$

This result has also been announced by Huber (1970) with indications of the proofs quite different from the above. Moreover, since $\text{Cov}(Y_i, Y_i^*) = 0$, it follows that M_n and S_n^2 are asymptotically independent.

We define the α, β -Winsorized mean as

$$(13) \quad W_n = n^{-1} \left[\sum_{i=r}^s X_{in} + (r-1)X_{r-1,n} + (n-s)X_{s+1,n} \right]$$

(See e.g. Bickel (1965)) and show that W_n can also be represented asymptotically as an average of i.i.d. random variables viz, if

$$(14) \quad \psi^{**}(x) = \begin{cases} (g_1 - \alpha/F'(g_1)) & \text{for } x < g_1 \\ x & \text{for } g_1 \leq x \leq g_2 \\ (g_2 + \beta/F'(g_2)) & \text{for } x > g_2 \end{cases}$$

and

$$(15) \quad Y_i^{**} = \psi^{**}(X_i) - E\psi^{**}(X_i)$$

then we prove

Theorem 3

$$(16) \quad W_n - \mu = n^{-1} \sum_{i=1}^n Y_i^{**} + o(n^{-3/4} \log n) \text{ a.s.,}$$

$$\text{where } \mu = \alpha g_1 + \beta g_2 + \int_{g_1}^{g_2} x dF(x).$$

Remarks similar to those following theorems 1 and 2 also apply here.

In section 2 we prove theorems 1, 2 and 3 and in section 3 we outline further applications to some sequential procedures based on M_n .

2. Proofs of the theorems

Proof of theorem 1

Denoting by $I(E)$ the indicator function of the event E , we can write

$$(17) \quad \sum_{i=1}^n X_{in} = \sum_{i=1}^n X_i I(g_1 \leq X_i \leq g_2) + V_{1n} + V_{2n}$$

where

$$(18) \quad V_{1n} = S_n (I(X_m < g_1) - I(X_m > g_1))$$

with S_n denoting the sum of those X_i 's in the interval $[g_1, X_m)$ if $g_1 < X_m$ and $[X_m, g_1)$ if $g_1 > X_m$. V_{2n} is similarly defined.

On the event $[X_m > g_1]$ the number of terms in $[g_1, X_m)$ equals $((r-1) - A_n)$ with A_n the number of X_i 's strictly less than g_1 . Hence on this event

$$((r-1) - A_n) g_1 \leq S_n \leq ((r-1) - A_n) X_m.$$

Similarly on the event $[X_m < g_1]$

we have

$$(A_n - (r-1)) X_m \leq S_n \leq (A_n - (r-1)) g_1$$

so that we have

$$(19) \quad (A_n - (r-1))X_{rn} \leq V_{1n} \leq (A_n - (r-1))g_1.$$

By the Bahadur representation of quantiles (See e.g. Bahadur (1966)) we find

$$(20) \quad X_{rn} = g_1 + n^{-1}((r-1) - A_n)/F'(g_1) + O(n^{-1/2}(\log \log n)^{1/2}) \text{ a.s.}$$

By the law of the iterated logarithm

$$(21) \quad A_n - (r-1) = O(n^{1/2}(\log \log n)^{1/2}) \text{ a.s.}$$

Hence

$$(22) \quad X_{rn} = g_1 + O(n^{-1/2}(\log \log n)^{1/2}) \text{ a.s.}$$

Applying (21) and (22) in (19) we find that

$$(23) \quad V_{1n} = (A_n - (r-1))g_1 + O(\log \log n) \text{ a.s.}$$

Similarly we find

$$(24) \quad V_{2n} = (B_n - (n-s))g_2 + O(\log \log n) \text{ a.s.}$$

with B_n the number of X_i 's strictly greater than g_2 .

Substituting (23) and (24) in (17) and rearranging terms we get

$$\begin{aligned} n\gamma M_n &= \sum_{i=1}^n X_i I(g_1 \leq X_i \leq g_2) + g_1 A_n + g_2 B_n - n\alpha g_1 - n\beta g_2 + O(\log \log n) \\ &= \sum_{i=1}^n Y_i + n\theta \gamma + O(\log \log n) \text{ a.s.} \end{aligned}$$

from which the result follows.

Remark

The regularity assumptions on F at g_1 and g_2 were required only in establishing (20); actually these assumptions yield much more than is required here namely that the remainder term is $O(n^{-3/4} \log n)$ (See Bahadur (1966)) or even smaller (See Kiefer (1967)). Thus any weaker regularity conditions which guarantee (20) would be sufficient

for our result. For example if $F'(g_1)$ and $F'(g_2)$ exist and are strictly positive then it is known that the remainder term in (20) is $o(n^{-1/2})$ in probability (See Ghosh (1971)) and consequently (5) holds in probability under this weaker assumption on F .

Proofs of theorem 2 and 3

These proofs are somewhat tedious but in principle quite analogous to the one given above. (Use $O(n^{-3/4} \log n)$ for the remainder terms in (20)).

3. Some applications to sequential procedures

3.1 A fixed width sequential confidence interval for the centre of symmetry of a symmetric distribution, similar to that of Chow and Robbins (1965), can be based on the trimmed mean rather than the mean. More specifically, if we define $L_n = M_n - S_n K_\alpha / n^{1/2}$ and $U_n = M_n + S_n K_\alpha / n^{1/2}$, where $\Phi(K_\alpha) = 1 - \alpha$ and Φ is the standard normal df, then we can define the stopping variable N to be the first integer $n \geq n_0$ (a fixed positive integer) such that $U_n - L_n \leq 2d$, with d a specified positive constant. The interval (L_N, U_N) forms the required fixed width sequential confidence interval. The asymptotic properties of this procedure are those given in theorem 2.1 of Geertsema (1970), since assumption 2.1 holds; this can be seen as follows. (i) and (ii) hold with $A = \sigma^{-1}$. Part (iii) follows from theorem 1. Part (iv) can be proved by substantially the same argument as used in the proof of lemma 3.3 of Geertsema (1970).

Asymptotic efficiencies of this procedure relative to other procedures can be defined as in section 5 of Geertsema (1970). Thus the asymptotic relative efficiency of the trimmed mean procedure compared to the Chow-Robbins procedure based on the (ordinary) mean is σ_0^2 / σ^2 where

$$(25) \quad \sigma_0^2 = \int_{-\infty}^{\infty} (x - \theta)^2 dF(x).$$

This is the same as the asymptotic relative efficiency in terms of asymptotic variances of the β, β -trimmed mean compared to the ordinary mean, and this can be quite large when F has heavy tails.

3.2 It is also possible to define a sequential selection procedure for the problem of selecting the population with the largest location parameter based on β, β -trimmed means similar to the ordinary means procedure of Robbins, Sobel and Starr (1968) and Srivastava (1966) for this procedure.

Asymptotic properties of such procedures similar to those given in these papers can be derived as above. Asymptotic relative efficiencies can be defined and discussed as in Swanepoel and Geertsema (1973).

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